A Project Tittle

On

Customer Segmentation Using K-Means and Hierarchical Clustering

Machine Learning

Bachelor of technology

In

A.Y.:2026-27

By

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DEPARTMENT OF CS&IT

Name of Title: Multi-Class Customer Category Prediction Using Multinomial Logistic Regression.

1.Abstract:

Customer segmentation is an essential task in e-commerce that helps businesses understand customer behavior and provide personalized services. This project, **"Multi-Class Customer Category Prediction Using Multinomial Logistic Regression,"** aims to classify customers into multiple categories based on their demographic information, purchasing behavior, and transaction history. Unlike binary classification, multinomial logistic regression is capable of predicting one among several possible customer categories, making it suitable for real-world marketing applications.

The proposed system uses customer-related features such as age, gender, income, purchase frequency, spending amount, and product preferences to train a multinomial logistic regression model. The dataset is preprocessed through data cleaning, handling missing values, feature encoding, and normalization to improve model performance. The trained model predicts the most probable customer category for each individual, enabling businesses to identify customer segments accurately.

This approach helps organizations design targeted marketing campaigns, improve customer satisfaction, increase sales, and optimize resource allocation. The model is evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix to measure its effectiveness. The results demonstrate that multinomial logistic regression provides an efficient, interpretable, and reliable solution for multi-class customer classification in e-commerce environments.

2. INTRODUCTION:

In today's competitive e-commerce environment, understanding customer behavior has become essential for businesses to improve customer satisfaction, increase sales, and develop effective marketing strategies. Customers differ in their purchasing habits, spending patterns, preferences, and demographics. Identifying these differences enables organizations to classify customers into meaningful categories and provide personalized products, services, and promotional offers.

Traditional customer classification methods often rely on manual analysis or simple binary classification techniques, which are insufficient when customers need to be grouped into multiple categories. To address this challenge, machine learning techniques have become increasingly popular for customer segmentation and classification. Among these techniques, Multinomial Logistic Regression is a supervised learning algorithm specifically designed for multi-class classification problems. It predicts the probability that a customer belongs to one of several predefined categories based on multiple input features.

The project, "Multi-Class Customer Category Prediction Using Multinomial Logistic Regression," focuses on developing a predictive model that classifies customers into different categories using attributes such as age, gender, annual income, purchase frequency, spending score, and buying behavior. The dataset undergoes preprocessing steps, including data cleaning, handling missing values, feature encoding, and normalization, to improve prediction accuracy. The trained model is then evaluated using standard performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

The proposed system assists businesses in identifying customer segments efficiently, enabling targeted marketing, personalized recommendations, improved customer relationship management, and better business decision-making. By leveraging multinomial logistic regression, the project provides an interpretable, accurate, and scalable solution for multi-class customer category prediction in modern e-commerce applications.

3. **SOFTWARE REQUIREMNETS**:

- **Operating System:** Windows 10/11 or Linux
- **Programming Language:** Python 3.x
- **Development Environment:** Jupyter Notebook or Visual Studio Code
- **Libraries Used:**
  - **NumPy** – Numerical computations
  - **Pandas** – Data manipulation and analysis
  - **Matplotlib** – Data visualization
  - **Seaborn** – Statistical data visualization
  - **Scikit-learn** – Implementation of Multinomial Logistic Regression, data preprocessing, model training, and evaluation
  - **Joblib** – Saving and loading the trained machine learning model
- **Dataset:** Customer Category Prediction Dataset (CSV format)

Signature of the Guide Course Instructor